

EXERCISE 1^{OR}**RESISTOR CODES - MEASURING INSTRUMENTS**

Resistors **are** used in electronic circuits to regulation, in general, of the current and are made of various materials and in different forms depending on the application. The most common types of resistors are the well-known ohmic carbon (graphite) and wire resistors.

Carbon resistors are manufactured for relatively low power (0.25 W - 3 W) and their values range from a few tenths of an Ω to about 50 M Ω . The resistance value is indicated on the component either with numbers or with special colors, as described in the color code of table 1 and in the example of figure 1. They are shown usually 4 colored stripes a, b, c and d, as shown in figure 1. The value of resistance is:

$$R (10^a \Omega) \times 10^b \quad (1)$$

while its tolerance is determined by d.

TABLE 1 Resistor color code

Color	1st strip	2nd strip	3rd lane (multiplier)	4th lane (tolerance)	Thermal coefficient
Black	0	0	$\times 100$		
Brown	1	1	$\times 101$	$\pm 1\%$ (F)	100 ppm
Red	2	2	$\times 102$	$\pm 2\%$ (G)	50 ppm
Orange	3	3	$\times 103$		15 ppm
Yellow	4	4	$\times 104$		25 ppm
Green	5	5	$\times 105$	$\pm 0.5\%$ (D)	
Blue	6	6	$\times 106$	$\pm 0.25\%$ (°C)	
Purple	7	7	$\times 107$	$\pm 0.1\%$ (B)	
Gray	8	8	$\times 108$	$\pm 0.05\%$ (A)	
White	9	9	$\times 109$		
Golden			$\times 0.1$	$\pm 5\%$ (J)	
Silver			$\times 0.01$	$\pm 10\%$ (K)	
None				$\pm 20\%$ (M)	

FIGURE 1

Carbon resistance color code

The accuracy of the value of each resistor varies within certain limits determined by the tolerance and are also indicated on the component either with numbers or with special colors. The usual tolerances are 10%, 5%, 2% or 1%. To understand the concept of tolerance, a simple example is given: resistance 100 Ω with a tolerance of 5%

means that its actual value can be from 95 to 105 Ohms. The best quality resistors are those with a tolerance of 1% or 0.5%. In the usual series of ohmic resistors of (E12) with a tolerance of 5% there are the following nominal resistance values: 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2 Ω , as well as all multiple values (e.g. 33 k Ω , 5.6 k Ω). In the resistor series with 2% or 1% tolerance (E24 and E96) other resistors are also available. values, but it should be noted that the resistance value is given with four colored strips a, b, c, d and there is an additional strip e for tolerance. The resistance value will be:

$$R = (100 \pm 10 \pm \gamma) \cdot 10^e \quad \gamma$$

while its tolerance is determined by e. There may also be a strip for the coefficient of temperature change.

Wirewound resistors are manufactured for higher powers 4 W - 1000 W and their values range from a few Ω to about 50 k Ω with tolerances of 5% or 10%.

The **capacitor** is a component consisting of two conductors (armatures) placed at a short distance from each other, between which air or other insulating material (dielectric) is inserted.

The **capacitance** of a capacitor is given by the formula:

$$C = Q / V \quad (3)$$

where Q is the charge on the capacitor and V is the voltage applied to its arms. The

The complex impedance it presents to a sinusoidal signal of frequency γ is:

$$Z = j(1/C\gamma) \quad (4)$$

The capacitance depends on: a) the size of the conductors (metal surfaces), b) the distance between them and c) the type of dielectric. Unit of measurement of capacitance is the Farad (F). However, it is very large in applications its multiples (mF, μ F, nF, pF) are used. The value of the capacitance is written on the component either in numbers (for example, 4n7 which means 4.7 nF), or with colors, as in resistors. The value with the color code is usually in pF.

There are many types of capacitors: paper, ceramic, polyester, electrolytic, variables etc. The basic characteristics of each type are the capacitance range (Capacitance range), the maximum voltage (maximum voltage), the accuracy of the value (Accuracy), the Temperature stability and leakage.

The multiples and submultiples of the various units of measurement have special names and symbols given in table 2.

Various semiconductor components will be used in the laboratory exercises. Figure 2 shows the terminals of some of them. Figure 3 shows the schematic representation of a connection board (mini-board or raster) to be used. In the figure one horizontal group of positions, which are electrically connected, appears shaded between them, as well as 4 (random) vertical groups of (five) positions. The horizontal groups are usually used for supply voltages and grounding.

TABLE 2

NAME	SYMBOL	POWER OF 10
Tera	T	10 ¹²
Giga	G	10 ⁹
Mega	M	10 ⁶
Kilo		10 ³
Hecto		10 ²
Deca	kh	10 ¹
Deci	da d	10 ⁻¹
Cents	c	10 ⁻²
Millie	m	10 ⁻³
Micro	m or μ	10 ⁻⁶
Nano	n	10 ⁻⁹
Pico		10 ⁻¹²
Femto	pf	10 ⁻¹⁵
Atto	a	10 ⁻¹⁸

FIGURE 2.1 Accessory Terminals

FIGURE 2.2 Accessory Terminals

FIGURE 3 Schematic representation of a connection board (raster)

FIGURE 4: Laboratory Test Chamber

EXPLANATION OF LABORATORY

1. ON-OFF switch
2. Voltage amplitude regulator in alternating voltage
3. Frequency multiplier in combination with button 4
4. Frequency adjustment selector in combination with button 3
5. AC voltage output waveform selector (sine, triangular, square)
6. Potentiometer 0~100k Ω
7. Potentiometer 0~1k Ω
8. DC voltage adjustment potentiometer 0 ~ 15V
- 8out.** 0 ~ 15V DC (adjustable by potentiometer 8)
9. DC voltage adjustment potentiometer 0 ~ -15V
- 9out.** 0 ~ -15V DC (adjustable by potentiometer 9)
10. Output +5V DC constant
11. 0V (ground)
12. Output -5V DC constant
13. AC voltage output

EXERCISE 2

PALMOGRAPH – VOLTAGE MEASUREMENT – FREQUENCY MEASUREMENT –

PHASE MEASUREMENT

THEORETICAL PART

The oscilloscope is an instrument for measuring voltage, frequency and phase. Also, through the oscilloscope we can observe various waveforms, transistor characteristics, response curves, proper operation of components, etc. The waveforms that appear on the

oscilloscope screen are formed by a moving bright spot on it. The trajectory of the spot on the screen is the result of two movements, one along the vertical axis and one along the horizontal axis. The vertical movement is due to an external voltage, while the horizontal movement is due to a voltage that is generated inside the oscilloscope by the so-called sweep generator, and whose waveform is that of Figure 1. The voltage that periodically and uniformly moves the spot from left to right on the oscilloscope screen and then very quickly from right to left is called a sawtooth sweep voltage.

FIGURE 1

In our laboratories we use the HAMEG HM203-5 oscilloscope, the front panel layout with the display and the various adjustment buttons is shown in Figure 2. In the following paragraphs we give a brief explanation of the symbols and functions of the FrontPanel.

FIGURE 2

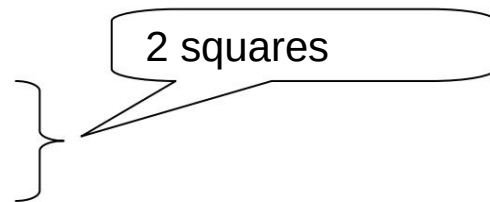
1. **POWER ON/OFF** : The oscilloscope's power switch.
2. **INTENS** : Adjusts the intensity of the beam on the screen.
3. **TR (TRIMMER)** : Rotates the beam to the center of the screen for trimming.
horizontalization.
4. **FOCUS** : Focuses the beam on the screen.
5. **Y-POS I, Y-POS II**: Adjusts the vertical position of the beam of channel I and II.
6. **Y-MAGx5** : In-out switch. In the (in) position, it magnifies the beam 5 times in y-axis.
7. **Y-MAGx5** : In-out switch. In the (in) position, it magnifies the beam 5 times in y-axis.
8. **Y-POS I, Y-POS II**: Adjusts the vertical position of the beam of channel I and II.
9. **SLOPE +/-**: In the outer position, triggering occurs on the rising part of the waveform. In the inner position, triggering occurs on the falling part.
10. **LEVEL** : In normal triggering (NORMAL TRIGGERING) it adjusts at what height the Waveform will begin to scandalize.
11. **X-POS** : Adjusts the position of the beam on the x-axis (right – left).
12. **X-MAGx10** : In – out switch. In the (in) position, it magnifies the beam 10 times in x-axis.
13. **VOLTS / DIV** : 12-position rotary switches from 5 mV / cm to 20 V / cm. The left one for the left channel and the right one for the right channel of the vertical amplifier.
14. **VOLTS / DIV** : Micrometric potentiometer for continuous amplitude variation. Increases the sensitivity of the vertical Y axis when rotated to the right. To obtain normal measurements, it must be turned to the CAL position all the way to the right.
15. **CH I / II – TRIG I / II**: In – out switch. Selects the operating channel I or II.
16. **DUAL** : In-out switch. Selects operation of one channel “in the out position” or both channels “in the in position”.
17. **ADD – CHOP** : In – out switch. Selects alternating or interrupted beams of operation. In the (out) position it gives the sum of the waveforms of the two channels.
18. **VOLTS / DIV** : 12-position rotary switches from 5 mV / cm to 20 V / cm. The left one for the left channel and the right one for the right channel of the vertical amplifier.
19. **VOLTS / DIV** : Micrometric potentiometer for continuous amplitude variation. Increases the sensitivity of the vertical Y axis when rotated to the right. To obtain normal measurements, it must be turned to the CAL position all the way to the right.
20. **TRIGGER SELECTOR** : Selects the trigger coupling AC, DC, HF, LF, and the 50 Hz coupling in the Line position.
21. **AT / NORM** : In the (out) position, triggering is automatic. In the (in) position, triggering is normal (NORMAL TRIGGERING).
22. **ALT** : Selects the trigger coupling between channels CH I and CHII with pressed the DUAL.
23. **HOLD OFF** : Sets the hold time between waveforms.
24. **TIME / DIV** : 18-position rotary switch for time base from 0.5 μ s / cm to 200 ms / cm.
25. **TIME / DIV** : Micrometric potentiometer for continuous variation of the time base. For normal measurements it must be turned to the CAL position all the way to the right.
26. **XY**: Selects the X – Y mode. In the (in) position, the time base circuit is deactivated and channel II is connected to input X.
27. **EXT** : In the (in) position selects external triggering.
28. **CH I (II)**: BNC connectors for signal input to channel I (left) and channel II (or horizontal x) (right). Right next to them is a separate ground input for banana plugs.
29. **CH. I, DC, AC, GD**: 3-position switch. Used to couple input I to **CH II, DC, AC, GD** and II (channel I, channel II). In the DC position, a direct connection is made. In the AC position, a connection is made through a capacitor. In the GD position, the input is grounded.
30. **GD** : Ground.
31. **(4mm socket)**: Ground.
32. **CH I (II)**: BNC connectors for signal input to channel I (left) and channel II (or horizontal x) (right). Right next to them is a separate ground input for terminals

banana type.

- 33. CH. I, DC, AC, GD:** 3-position switch. Used to couple input I to **CH II, DC, AC, GD** and II (channel I, channel II). In the DC position, a direct connection is made. In the AC position, a connection is made through a capacitor. In the GD position, the input is grounded.
- 34. GD :** Grounding.
- 35. INV I:** In-out switch. Inverts channel I and II.

- 36. TRIG INP :** This is a BNC connector, input for an external trigger source. It is used in conjunction with the EXT switch.
- 37. COMPONENT TESTER :** In-out switch and single-pole plug. The measurement is made by connecting one cable to the single-pole plug and one to one of the two grounds of the oscilloscope. The illustrations of the various materials are shown in Figure 3.
- 39. CAL 0.2 V, 2 V:** Two square pulse output pins of 0.2 V height, the upper one and 2 V the bottom.
- 40. CALIBRATOR 1kHz/1MHz:** Micrometrically calibrates the oscilloscope scale. Switch
Out: γ 1kHz Switch In: γ
1MHz

FIGURE 3

TYPES OF MEASUREMENTS WITH PULSOGRAPH1. DC VOLTAGE MEASUREMENT .

To measure the DC voltage that we apply to the vertical deflection of the oscilloscope, we measure the point from its original position when we apply the voltage in the figure above. For example, if the deviation of the point is two squares and the Volts / DIV button is in the 2 Volts position, the voltage we measure is $2 \times 2 = 4$ V

FIGURE 42. AC VOLTAGE MEASUREMENT

There are three ways to define the measure of alternating voltage:

V_{pp} : peak to peak, which means that the wave length is measured from one end to the other. So by applying an alternating voltage to the input of the vertical deflection of the oscilloscope, we measure its height, that is, how many squares it covers and multiply by the position

FIGURE 5

of the Volts / DIV switch. For example, if the height of the measured voltage covers 4 squares, and the Volts / DIV switch is in the 1 Volt position, the measured voltage has

value equal to $4 \times 1 = 4V_{pp}$.

V_p **peak p:** the wave modulus is measured from the wave's axis of symmetry to one of its ends. This value is numerically equal to half of V_{PP} , that is

applies: $V_p = \frac{V_{pp}}{2}$.

V_{eff} **effective value:** for a sinusoidal voltage the following applies: $V_{eff} = \frac{V_p}{\sqrt{2}}$.

3. FREQUENCY MEASUREMENT

After connecting the alternating voltage to the vertical deviation, we place the Time / DIV switch in such a position that we get a sine wave shape on the screen (see figure 5). At this point we should clarify two definitions: a) **Cycle (C)**: it is the part of the wave whose multiples constitute a

waveform.

b) **Period (T):** the amount of time in seconds required by the wave to completing a full cycle.

c) **Frequency (F):** is the number of cycles in one sec or cycles per second (c/sec) and is measured in Hz. The mathematical relationship that connects the period to the

frequency is given by the equation: $F = \frac{1}{T}$

where F is the frequency in Hz and T is the period in sec.

To measure the period of a specific waveform, the horizontal axis of oscilloscope, called the x-axis, is graduated in time units sec / cm (Time / cm). The period is therefore calculated from the oscilloscope screen as follows: for example in figure 5, assuming the Time/DIV knob is in position 1 μ sec, then $T = 4 \text{ squares} \times 1 = 4 \mu\text{sec}$. From the above frequency formula, we have that:

$$F = \frac{1}{T} = \frac{1}{4 \times 10^{-6}} = 250000 \text{ Hz} = 250 \text{ KHz}$$

4. PHASE DIFFERENCE MEASUREMENT

If two waves of the same frequency reach their maximum at different times, we say that there is a phase difference between them and that one precedes the other by an angle ϕ .

The phase difference (angle ϕ) can be observed and measured on an oscilloscope with double signal beams. The two equal-frequency signals are connected to the two axes. The figure

6 shown next to shows two waves with phase difference ϕ .

There are two ways to find

The first way is as follows:

First we count how many squares is the distance 2 peaks of 1 of waveform.

From figure 6 we see that

FIGURE 6

is 4 squares. Then we count how many squares apart the two vertices of two waveforms (1) and (2), which from figure 6 we see is a square.

When making the measurement it is necessary to set equal deviation on both axes adjusting the horizontal and vertical amplifiers respectively. So after the previous measurements we continue the calculation with a simple method of three, as

following:

$$\begin{array}{l} \text{A (4 Div)} \\ \text{B (1st Division)} \end{array} \quad \begin{array}{l} = \\ = \end{array} \quad 360^\circ ; = \phi$$

$$\text{So } \phi = x = 360^\circ \times \frac{1}{4} \rightarrow \phi = 90^\circ .$$

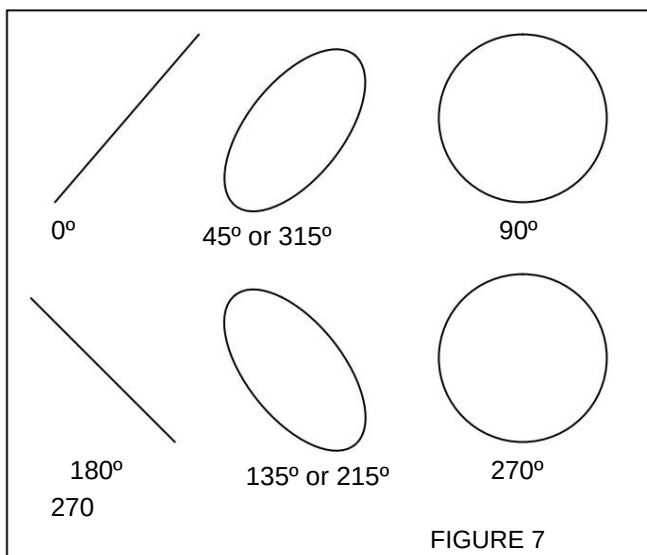


FIGURE 7

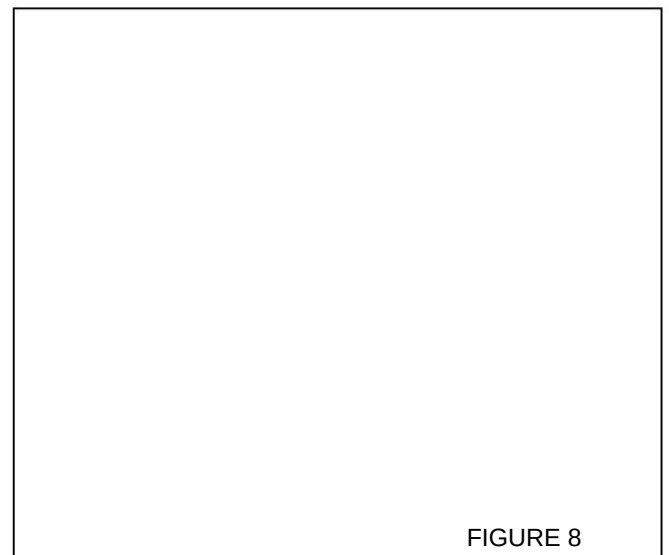


FIGURE 8

The second way to measure the phase difference is as follows: The two equal-frequency signals are again connected to the two axes. And while we observe the two waveforms on the oscilloscope screen as shown in figure 6, we press the (X-Y) button on the oscilloscope (attention, with the Dual button out) and we observe that one of the shapes in figure 7 is formed on the oscilloscope screen, namely a circle, an ellipse or a straight line. When making the measurement, it is necessary to adjust equal deviation on the two axes, by adjusting the horizontal and vertical amplifier. Figure 8 shows an oscilloscope representation caused by a phase difference between the two input signals. An ellipse with a slope to the right is presented for a phase difference from 0° to 90° . So with the help of figure 8, the calculation of the phase difference is done in the following ways:

$$\phi = \sin^{-1} \left(\frac{y}{x} \right)$$

where: ϕ = phase angle in degrees

y = maximum value of ellipse in horizontal direction in cm

x = value at the intersection of the ellipse and the horizontal axis in cm.

LABORATORY PART

1. DC voltage measurement: Measure the +5volt DC voltage of your test bench using your oscilloscope and multimeter. Note down all the readings of the instruments and the calculations you make in detail on the board.

<u>Oscilloscope Display</u>	<u>Voltmeter Display</u>

2. AC voltage measurement

- Set the signal generator to a frequency of 1000 Hz and the output to 1 Volt (active value).
- Set the sensitivity of the (Y) axis on the oscilloscope to 1 Volt / cm.
- Connect the signal generator to the (Y) axis input of the oscilloscope.
- Set the time selector switch to show a few cycles on the screen.
- We measure the maximum value as well as the peak value, calculate the effective value and We record the results in Table 1.
- We repeat the above for signals with an effective value of 2 and 2.5 (Note: We measure the effective value (V_{eff}) from the generator output with the help of a voltmeter).

Input signal active value in Volts	Voltage value V_{p-p} (peak to peak) in Volts	Calculation of voltage value V_p in Volts $V_p = V_{p-p} / 2$	Calculation of effective value V_{eff} in Volts $V = V_p / \sqrt{2}$
1			
2			
2.5			

TABLE 1

3. Measure the period and calculate the frequency F from it. i. Set the signal generator to a frequency of 1000 Hz and the output to 5

V_{pp} .

ii. Connect the output of the signal generator to the vertical deviation (Y) of the oscilloscope.

iii. We calculate the period (T) of the waveform, as well as the frequency that corresponds and we record the results in Table 2. iv. We repeat the above for frequencies 2000, 3000, 4000 and 5000 Hz.

Frequency f/Hz	Period T/msec	Frequency calculation Hz
1000		
2000		
3000		
4000		
5000		

TABLE 2

4. Measurement of the phase

difference. i. We connect the following circuit:

CIRCUIT

ii. Set the signal generator to a frequency of 500 Hz and an output of 5 V_{pp} maximum value voltage.

iii. We adjust the oscilloscope for equal deviations on the axes. iv.

From the waveform that appears on the oscilloscope screen, we measure the distances $\dot{y}$ and $\ddot{y}$ and record the values in Table 3. We calculate the phase difference $\dot{y}$ with the formulas from the second method mentioned in the theory.

v. (Note: The (X-Y) button is "middle")

Without changing the frequency and with the (X-Y) button pressed in the "out" position, we calculate the phase difference of the two waveforms as mentioned in the first calculation method in the theory. We also enter the values in Table 3 on the next page.

Frequency Hz	A (div)	B (div)	Phase difference (simple method of three) (ϕ°)	a (div)	ϕ (div)	Phase difference ($\arcsin(a/b)$) (ϕ°)
500						
1000						
2000						
3000						
4000						
5000						

TABLE 3**QUESTIONS**

1. What is the value of the DC voltage that causes the position to deviate by 3 squares?
and the Volt / DIV switch is in position 2?
2. What are the values of the alternating voltage V_p , V_p and V_{ϕ} , which causes the dot to deviate by 0.5, 2 and 3 squares respectively from peak to peak and the Volt / DIV switch is set to 50 mV, 2 V and 5 V respectively?
3. What is the frequency of the alternating voltage when the peak-to-peak deviation is 0.5, 2 and 3 squares and the Time / DIV switch is in the 2, 5 and 50 ms position respectively?
4. What is the phase difference of two alternating voltage waveforms when they are 1.5 squares apart in time and the distance between two consecutive peaks of one waveform is 6 squares?
5. Calculate the theoretically expected values of the phase difference ϕ using the formula: $\phi = \arctan(1 / (2\pi f RC))$ and the 6 frequency values from table 3 knowing that $f = 3.14$ f is the frequency of the alternating voltage for which the measurement was made, $R = 1k\Omega$ is the resistance used to construct the circuit and $C = 0.1\mu F$ is the capacitance used to construct the circuit. Then compare the theoretical values with those in table 3.